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CUET UG

Mathematics

Integration and its Applications

Indefinite & Definite Integrals, Properties, Leibniz Rule, and Area Bounded by Curves

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Chapter Overview: The Integral Calculus Engine

Integration represents one of the dual pillars of infinitesimal calculus, serving both as the inverse operation to differentiation (the **antiderivative**) and as the rigorous foundational mechanism for computing continuous accumulations, total displacements, and geometrically bounded areas. In the CUET UG Mathematics examination, Integral Calculus constitutes between **18% to 22% of total exam weightage**, spanning both Section A (compulsory common core) and Section B1 (core mathematics).

A mastery of integration demands far more than mechanical memorization of antiderivative tables. High-scoring aspirants must master three interconnected intellectual capabilities:

1. **Pattern Recognition in Indefinite Integrals:** Instantly diagnosing whether an integrand yields to direct algebraic manipulation, trigonometric substitution, integration by parts via the ILATE hierarchy, partial fraction decomposition, or one of the 9 special quadratic forms.
2. **Definite Integral Symmetries & The King's Rule:** Utilizing definite integral properties (P_0 through P_7) to collapse intractable integrands into trivial constants without ever finding the indefinite antiderivative.
3. **Geometric Precision in Area Bounded (AOI):** Accurately sketching bounding curves, locating intersections, choosing the optimal axis of slicing (vertical strips dx vs horizontal strips dy), and deploying high-speed competitive shortcuts for parabolas, ellipses, and trigonometric loops.

THE FUNDAMENTAL DUALITY OF CALCULUS

Let $f : [a, b] \to \mathbb{R}$ be a continuous real-valued function:

- **The Indefinite Integral:** $\int f(x)dx = F(x) + C$, where $F'(x) = f(x)$ and $C \in \mathbb{R}$ is an arbitrary constant of integration representing an entire one-parameter family of mutually parallel curves.
- **The First Fundamental Theorem of Calculus (FTC 1):** If f is continuous on $[a, b]$ and $A(x) = \int_a^x f(t)dt$, then $A'(x) = f(x)$ for all $x \in [a, b]$.
- **The Second Fundamental Theorem of Calculus (FTC 2):** If $F'(x) = f(x)$ on $[a, b]$, then $\int_a^b f(x)dx = [F(x)]_a^b = F(b) - F(a)$. The arbitrary constant C identically cancels out.

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1. Indefinite Integration: Methods & Core Forms

An indefinite integral represents the totality of all antiderivatives of a given function. If $F(x)$ satisfies $\frac{d}{dx}[F(x)] = f(x)$, then:

$$\int f(x)dx = F(x) + C$$

Geometrically, $y = F(x) + C$ represents a family of congruent curves obtained by translating the graph of $y = F(x)$ vertically along the y-axis. The tangents to all these curves at any ordinate $x = x_0$ are strictly parallel, each possessing identical slope $m = f(x_0)$.

1. Standard Elementary Antiderivatives

Integrand Family	Indefinite Integral Formula	Domain Restrictions / Notes
Power Functions	$\int x^n dx = \frac{x^{n+1}}{n+1} + C$	$n \neq -1$. Add $+1$ to power.

Integrand Family	Indefinite Integral Formula	Domain Restrictions / Notes
Reciprocal Function	$\int \frac{1}{x}dx = \ln\|x\| + C$	$x \neq 0$. Absolute value mandatory.
Natural Exponential	$\int e^{kx}dx = \frac{e^{kx}}{k} + C$	$k \neq 0$. Linear argument scaling.
General Exponential	$\int a^x dx = \frac{a^x}{\ln a} + C$	$a > 0, a \neq 1$. Divide by $\ln a$.
Sine Function	$\int \sin(kx)dx = -\frac{\cos(kx)}{k} + C$	Note the negative sign!
Cosine Function	$\int \cos(kx)dx = \frac{\sin(kx)}{k} + C$	Standard positive antiderivative.
Secant Squared	$\int \sec^2 x dx = \tan x + C$	From $\frac{d}{dx}[\tan x] = \sec^2 x$.
Cosecant Squared	$\int \csc^2 x dx = -\cot x + C$	From $\frac{d}{dx}[\cot x] = -\csc^2 x$.
Secant Tangent	$\int \sec x \tan x dx = \sec x + C$	Direct trigonometric derivative.
Cosecant Cotangent	$\int \csc x \cot x dx = -\csc x + C$	Negative trigonometric derivative.
Tangent Function	$\int \tan x dx = \ln\|\sec x\| + C = -\ln\|\cos x\| + C$	Derived via $\sin \frac{x}{\cos} x$ substitution.
Cotangent Function	$\int \cot x dx = \ln\|\sin x\| + C$	Derived via $\cos \frac{x}{\sin} x$ substitution.
Secant Function	$\int \sec x dx = \ln\|\sec x + \tan x\| + C$	Alternative: $\ln\|\tan(\frac{\pi}{4} + \frac{x}{2})\| + C$.
Cosecant Function	$\int \csc x dx = \ln\|\csc x - \cot x\| + C$	Alternative: $\ln\|\tan(\frac{x}{2})\| + C$.

2. Integration by Substitution (Change of Variable)

When an integrand cannot be integrated directly in its initial variable x, we transform the variable to t using an invertible substitution $x = g(t)$, giving $dx = g'(t)dt$:

$$\int f(g(x))g'(x)dx = \int f(t)dt \quad \text{where } t = g(x)$$

High-Yield Substitution Archetypes in CUET:

1. **The Golden Derivative-over-Function Form:**

$$\int \frac{f'(x)}{f(x)}dx = \ln|f(x)| + C$$

Example: $\int \frac{2x+3}{x^2+3x+7}dx = \ln|x^2 + 3x + 7| + C$.

2. **The Power Function Derivative Form:**

$$\int [f(x)]^n f'(x)dx = \frac{[f(x)]^{n+1}}{n+1} + C \quad (n \neq -1)$$

Example: $\int \sin^4 x \cos x dx = \frac{\sin^5 x}{5} + C$.

3. **Trigonometric Integrands with Odd/Even Powers:**
   - If power of $\sin x$ is odd, substitute $t = \cos x$.
   - If power of $\cos x$ is odd, substitute $t = \sin x$.
   - If powers of both $\sin x$ and $\cos x$ are even, convert using half-angle identities:

$$\sin^2 x = \frac{1 - \cos 2x}{2}, \quad \cos^2 x = \frac{1 + \cos 2x}{2}$$

STANDARD ALGEBRAIC & TRIGONOMETRIC SUBSTITUTIONS

When encountering algebraic radical expressions in the integrand:

- For $\sqrt{a^2 - x^2}$: Substitute $x = a\sin\theta$ or $x = a\cos\theta$ (since $1 - \sin^2\theta = \cos^2\theta$).
- For $\sqrt{a^2 + x^2}$ or $a^2 + x^2$: Substitute $x = a\tan\theta$ or $x = a\sinh\theta$ (since $1 + \tan^2\theta = \sec^2\theta$).
- For $\sqrt{x^2 - a^2}$: Substitute $x = a\sec\theta$ (since $\sec^2\theta - 1 = \tan^2\theta$).
- For $\sqrt{\frac{a-x}{a+x}}$: Substitute $x = a\cos 2\theta$.

2. The Special 9 Quadratic Integrals

Integrals involving quadratic expressions in the denominator or numerator form a cornerstone of the CUET syllabus. Mastering their exact functional forms and avoiding common coefficient errors is mandatory.

The Three Non-Radical Rational Denominators

1. $$\int \frac{dx}{x^2 - a^2} = \frac{1}{2a}\ln\left|\frac{x-a}{x+a}\right| + C$$
2. $$\int \frac{dx}{a^2 - x^2} = \frac{1}{2a}\ln\left|\frac{a+x}{a-x}\right| + C$$
3. $$\int \frac{dx}{x^2 + a^2} = \frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right) + C$$

The Three Radical Denominators

4. $$\int \frac{dx}{\sqrt{a^2 - x^2}} = \sin^{-1}\left(\frac{x}{a}\right) + C \quad \left(\textbf{CRITICAL: No } \frac{1}{a} \textbf{ factor outside!}\right)$$
5. $$\int \frac{dx}{\sqrt{x^2 - a^2}} = \ln|x + \sqrt{x^2 - a^2}| + C$$
6. $$\int \frac{dx}{\sqrt{x^2 + a^2}} = \ln|x + \sqrt{x^2 + a^2}| + C$$

The Three Radical Numerators (The AOI Triad)

7. $$\int \sqrt{a^2 - x^2}\,dx = \frac{x}{2}\sqrt{a^2 - x^2} + \frac{a^2}{2}\sin^{-1}\left(\frac{x}{a}\right) + C$$
8. $$\int \sqrt{x^2 - a^2}\,dx = \frac{x}{2}\sqrt{x^2 - a^2} - \frac{a^2}{2}\ln|x + \sqrt{x^2 - a^2}| + C$$
9. $$\int \sqrt{x^2 + a^2}\,dx = \frac{x}{2}\sqrt{x^2 + a^2} + \frac{a^2}{2}\ln|x + \sqrt{x^2 + a^2}| + C$$

LINEAR OVER QUADRATIC DECOMPOSITION ALGORITHM

To evaluate integrals of the form $\int \frac{px+q}{ax^2+bx+c}dx$ or $\int \frac{px+q}{\sqrt{ax^2+bx+c}}dx$:

1. Express the numerator in terms of the derivative of the quadratic radicand:

$$px + q = A\frac{d}{dx}[ax^2 + bx + c] + B = A(2ax + b) + B$$

2. Equate coefficients of x and constants to solve for A and B:

$$A = \frac{p}{2a}, \quad B = q - \frac{pb}{2a}$$

3. Split the integral into two distinct parts:

$$\int \frac{px+q}{ax^2+bx+c}dx = A\underbrace{\int \frac{2ax+b}{ax^2+bx+c}dx}_{\ln|\text{quadratic}|} + B\underbrace{\int \frac{dx}{ax^2+bx+c}dx}_{\text{Complete the square}}$$

3. Integration by Parts & Partial Fractions

1. Integration by Parts & The ILATE Protocol

Derived directly from the product rule of differentiation $\frac{d}{dx}[uv] = uv' + vu'$, the formula for integration by parts is:

$$\int u(x)v'(x)dx = u(x)v(x) - \int u'(x)v(x)dx$$

Or in standard student shorthand:

$$\int (\textbf{First}) \cdot (\textbf{Second})dx = (\textbf{First})\int (\textbf{Second})dx - \int \left[\frac{d}{dx}(\textbf{First}) \cdot \left(\int (\textbf{Second})dx\right)\right]dx$$

THE ILATE PRIORITY HIERARCHY

To select which factor serves as the **First Function $u(x)$** (the function to differentiate) versus the **Second Function $v'(x)$** (the function to integrate), follow the strict priority order:

- **I — Inverse Trigonometric Functions:** $\sin^{-1}x, \cos^{-1}x, \tan^{-1}x$ (Highest priority for u)
- **L — Logarithmic Functions:** $\ln x, \log_a x$
- **A — Algebraic Functions:** $x^n, x^2+1, \sqrt{x}$
- **T — Trigonometric Functions:** $\sin x, \cos x, \sec^2 x$
- **E — Exponential Functions:** e^x, a^x (Lowest priority for u, ideal second function)

Crucial Rule: When integrating isolated inverse trigonometric or logarithmic functions (e.g. $\int \ln x dx$ or $\int \tan^{-1}x dx$), introduce unity $1 = x^0$ as the algebraic second function!

$$\int \ln x dx = \int \underbrace{(\ln x)}_{u} \cdot \underbrace{1}_{v'}dx = (\ln x)x - \int \left(\frac{1}{x} \cdot x\right)dx = x\ln x - x + C = x(\ln x - 1) + C$$

2. The Exponential Master Shortcut: $\int e^x[f(x) + f'(x)]dx$

A standard, exceptionally frequent question type in CUET involves the product of an exponential function and a sum of terms:

$$\int e^x[f(x) + f'(x)]dx = e^x f(x) + C$$

Verification: By the product rule, $\frac{d}{dx}[e^x f(x)] = e^x f(x) + e^x f'(x) = e^x[f(x) + f'(x)]$.

- **Classic Example 1:** $\int e^x[\tan x + \sec^2 x]dx = e^x \tan x + C$, since $f(x) = \tan x \implies f'(x) = \sec^2 x$.
- **Classic Example 2:** $\int e^x\left[\frac{x-1}{x^2}\right]dx = \int e^x\left[\frac{1}{x} - \frac{1}{x^2}\right]dx = \frac{e^x}{x} + C$, since $f(x) = \frac{1}{x} \implies f'(x) = -\frac{1}{x^2}$.
- **General Form:** $\int e^{kx}[kf(x) + f'(x)]dx = e^{kx}f(x) + C$.

3. Rational Functions and Partial Fractions

When integrating a rational function $\frac{P(x)}{Q(x)}$:

1. **Proper Fraction Verification:** Verify that $\deg(P) < \deg(Q)$. If $\deg(P) \geq \deg(Q)$, perform polynomial long division first to express $\frac{P(x)}{Q(x)} = T(x) + \frac{R(x)}{Q(x)}$ where $\deg(R) < \deg(Q)$.
2. **Linear Non-Repeated Factors:**

$$\frac{px+q}{(x-a)(x-b)} = \frac{A}{x-a} + \frac{B}{x-b}$$

   Cover-Up Method: $A = \left[\frac{px+q}{x-b}\right]_{x=a}$ and $B = \left[\frac{px+q}{x-a}\right]_{x=b}$.
3. **Repeated Linear Factors:**

$$\frac{px^2+qx+r}{(x-a)^2(x-b)} = \frac{A}{x-a} + \frac{B}{(x-a)^2} + \frac{C}{x-b}$$

4. **Irreducible Quadratic Factor:**

$$\frac{px^2+qx+r}{(x-a)(x^2+bx+c)} = \frac{A}{x-a} + \frac{Bx+C}{x^2+bx+c}$$

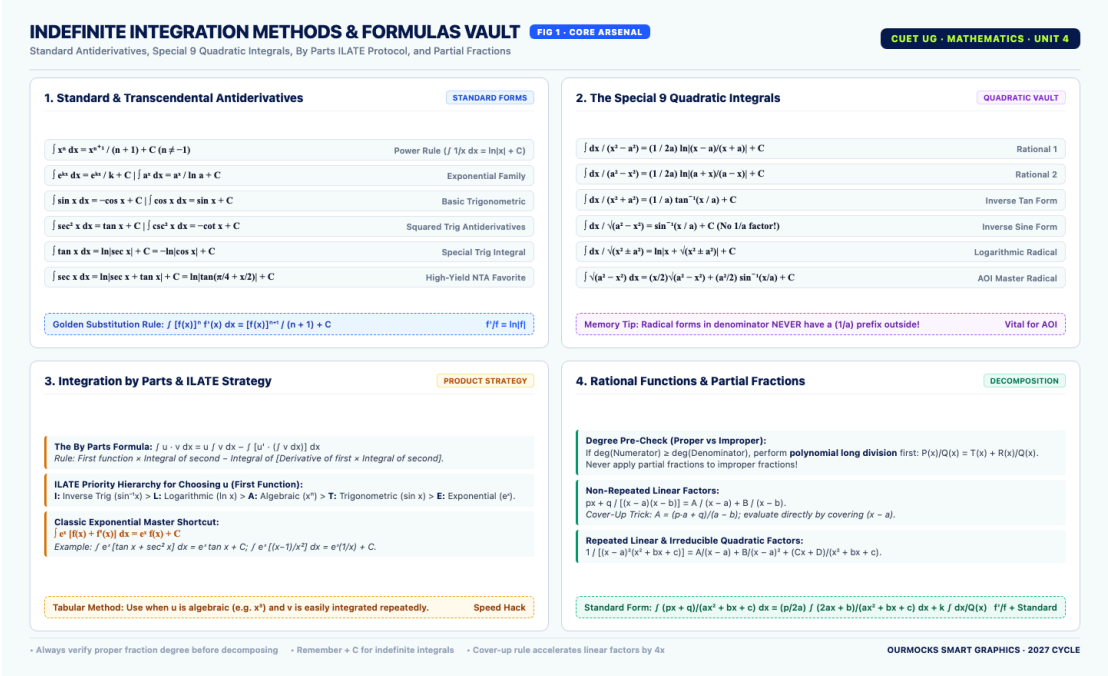


Figure 1: Figure 1: Comprehensive Indefinite Integration Methods, Special 9 Forms, ILATE Strategy, and Partial Fraction Decompositions.

Key Analytical Takeaways from Figure 1:

1. **The Three Radical Denominators Contrast:** Observe the critical structural difference between $\int \frac{dx}{\sqrt{a^2-x^2}} = \sin^{-1}\left(\frac{x}{a}\right) + C$ (which has NO $\frac{1}{a}$ factor outside) versus $\int \frac{dx}{x^2+a^2} = \frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right) + C$ (which possesses a mandatory $\frac{1}{a}$ prefix). NTA frequently inserts answer options containing an erroneous $\frac{1}{a}$ prefix on inverse sine integrals.
2. **Proper Fraction Prerequisite:** Attempting partial fractions on improper fractions without long division produces fatal algebraic contradictions.
3. **Cover-Up Acceleration:** For linear factors, the Heaviside cover-up method computes partial fraction coefficients in under 5 seconds, bypassing simultaneous equations.

COVER-UP METHOD VS SIMULTANEOUS LINEAR EQUATIONS

Problem: Evaluate the indefinite integral $\int \frac{2x+1}{(x-1)(x+2)(x-3)}dx$.

- **Standard Decomposition:** $\frac{2x+1}{(x-1)(x+2)(x-3)} = \frac{A}{x-1} + \frac{B}{x+2} + \frac{C}{x-3}$.
- **Heaviside Cover-Up Speed Technique:**
  - For A (set $x = 1$, cover $(x-1)$): $A = \frac{2(1)+1}{(1+2)(1-3)} = \frac{3}{(3)(-2)} = -\frac{1}{2}$.
  - For B (set $x = -2$, cover $(x+2)$): $B = \frac{2(-2)+1}{(-2-1)(-2-3)} = \frac{-3}{(-3)(-5)} = -\frac{1}{5}$.
  - For C (set $x = 3$, cover $(x-3)$): $C = \frac{2(3)+1}{(3-1)(3+2)} = \frac{7}{(2)(5)} = \frac{7}{10}$.
- **Antiderivative:**

$$\int \frac{2x+1}{(x-1)(x+2)(x-3)}dx = -\frac{1}{2}\ln|x-1| - \frac{1}{5}\ln|x+2| + \frac{7}{10}\ln|x-3| + C$$

CUET Speed Impact: Solves the partial fraction decomposition in 20 seconds flat without setting up or solving a 3×3 linear system!

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4. Definite Integrals: The Master Properties (P_0 to P_7)

A definite integral computes the net signed accumulation of a continuous function across a closed interval:

$$\int_a^b f(x)dx = \lim_{\{n\to\infty\}} h \sum_{\{r=1\}}^{n} f(a+rh) \quad \text{where } h = \frac{b-a}{n}$$

In CUET examination problems, explicit calculation of limits of Riemann sums is rarely required; instead, the vast majority of problems test deep mastery of the **8 Fundamental Definite Integral Properties**.

The 8 Canonical Properties of Definite Integrals

Property	Mathematical Statement	Exam Application / Utility
P_0	$\int_a^b f(x)dx = \int_a^b f(t)dt$	Dummy variable invariance.
P_1	$\int_a^b f(x)dx = -\int_b^a f(x)dx$	Limit interchange negates the integral; $\int_a^a = 0$.
P_2	$\int_a^b f(x)dx = \int_a^c f(x)dx + \int_c^b f(x)dx$	Essential for modulus $\lvert x-c\rvert$ & step $[x]$ functions.
P_3	$\int_a^b f(x)dx = \int_a^b f(a+b-x)dx$	General King's Property: Reflection across midpoint.
P_4	$\int_0^a f(x)dx = \int_0^a f(a-x)dx$	Standard King's Property: The ultimate CUET weapon.
P_5	$\int_0^{2a} f(x)dx = \int_0^a f(x)dx + \int_0^a f(2a-x)dx$	Interval halving decomposition.
P_6	$\int_0^{2a} f(x)dx = \begin{cases} 2\int_0^a f(x)dx & \text{if } f(2a-x)=f(x) \\ 0 & \text{if } f(2a-x)=-f(x) \end{cases}$	Queen's Property: Slices upper limit in half.
P_7	$\int_{-a}^a f(x)dx = \begin{cases} 2\int_0^a f(x)dx & \text{if } f(-x)=f(x) \quad \textbf{(Even)} \\ 0 & \text{if } f(-x)=-f(x) \quad \textbf{(Odd)} \end{cases}$	Symmetric limits: Odd functions vanish identically!

Deep Dive: The King's Property (P_4) Algorithm

The King's Property $\int_0^a f(x)dx = \int_0^a f(a-x)dx$ is responsible for solving over 60% of all definite integral problems in competitive examinations.

THE KING'S PROPERTY SYMMETRY COLLAPSE ALGORITHM

Whenever an integrand contains complementary trigonometric or exponential pairs on $[0,a]$ (such as $\sin x$ and $\cos x$ on $[0,\frac{\pi}{2}]$, or $f(x)$ and $f(a-x)$):

1. Label the original integral:
$$I = \int_0^a \frac{g(x)}{g(x)+g(a-x)}dx$$
2. Apply Property P_4 ($x \to a-x$):
$$I = \int_0^a \frac{g(a-x)}{g(a-x)+g(x)}dx$$
3. Add the two equations ($I + I = 2I$):
$$2I = \int_0^a \frac{g(x)+g(a-x)}{g(x)+g(a-x)}dx = \int_0^a 1dx = [x]_0^a = a$$

4. Solve for I:

$$I = \frac{a}{2}$$

Standard Classical Results via King's Rule:

- $\int_0^{\frac{\pi}{2}} \frac{\sin^n x}{\sin^n x + \cos^n x} dx = \frac{\pi}{4}$ for ANY real exponent n.
- $\int_0^{\frac{\pi}{2}} \frac{\sqrt{\tan x}}{\sqrt{\tan x} + \sqrt{\cot x}} dx = \frac{\pi}{4}$.
- $\int_0^{\frac{\pi}{2}} \ln(\sin x) dx = \int_0^{\frac{\pi}{2}} \ln(\cos x) dx = -\frac{\pi}{2} \ln 2$.

The Leibniz Rule of Differentiation Under the Integral Sign

When evaluating limits of indeterminate forms ($\frac{0}{0}$ or $\frac{\infty}{\infty}$) where integrals appear in the numerator, L'Hôpital's Rule requires differentiating the integral. The Leibniz Rule gives:

$$\frac{d}{dx}\left[\int_{u(x)}^{v(x)} f(t)dt\right] = f(v(x)) \cdot v'(x) - f(u(x)) \cdot u'(x)$$

Example: To find $\lim_{x\to 0} \frac{\int_0^{x^2} \cos(t^2)dt}{x \sin x}$:

- Direct substitution yields $\frac{0}{0}$. Applying L'Hôpital's Rule:

$$\text{Numerator derivative} = \cos\left((x^2)^2\right) \cdot \frac{d}{dx}[x^2] - 0 = 2x\cos(x^4)$$

$$\text{Denominator derivative} = \frac{d}{dx}[x \sin x] = \sin x + x \cos x$$

- Taking the limit:

$$\lim_{x\to 0} \frac{2x\cos(x^4)}{\sin x + x\cos x} = \lim_{x\to 0} \frac{2\cos(x^4)}{\frac{\sin x}{x} + \cos x} = \frac{2(1)}{1+1} = 1$$

5. Applications of Integrals: Bounded Area (AOI)

The Application of Integrals (AOI) in CUET tests geometric comprehension, spatial visualization of curves, and precise definite integration to evaluate the surface area of regions bounded by lines, parabolas, circles, ellipses, and trigonometric functions.

1. Fundamental Theorems of Bounded Area

1. **Area Bounded by $y = f(x)$, the x-axis, and Ordinates $x = a, x = b$:**

$$\text{Area} = \int_a^b |y|\ dx = \int_a^b |f(x)|\ dx$$

If the curve crosses the x-axis at $x = c \in (a, b)$, such that $f(x) \geq 0$ on $[a, c]$ and $f(x) \leq 0$ on $[c, b]$:

$$\text{Area} = \int_a^c f(x)dx + |\int_c^b f(x)dx| = \int_a^c f(x)dx - \int_c^b f(x)dx$$

Warning: Never compute $\int_a^b f(x)dx$ without modulus splitting! Net signed area allows negative lobes to cancel positive lobes, yielding an erroneous total area (e.g. $\int_0^{2\pi} \sin x dx = 0$, whereas the bounded area is 4 sq units).

2. **Area Bounded by $x = g(y)$, the y-axis, and Abscissae $y = c, y = d$:**

$$\text{Area} = \int_c^d |x| \, dy = \int_c^d |g(y)| \, dy$$

Horizontal slicing is vastly superior when curves are explicitly solved for x in terms of y (e.g. $x = y^2$).

3. **Area Between Two Curves $y = f(x)$ and $y = g(x)$ from $x = a$ to $x = b$:**

$$\text{Area} = \int_a^b \left[y_{\text{upper}} - y_{\text{lower}} \right] dx = \int_a^b [f(x) - g(x)] dx \quad \text{where } f(x) \geq g(x)$$

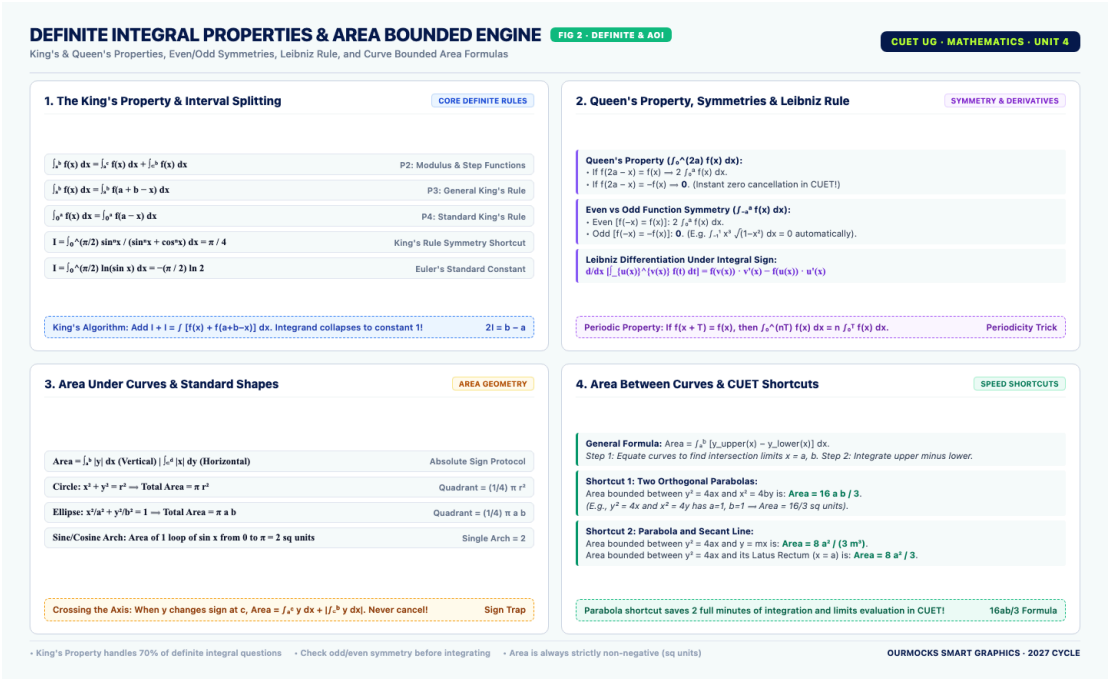


Figure 2: Figure 2: Definite Integral Properties, Symmetry & Leibniz Rule, and Area Bounded by Curves Geometric Engine.

Key Analytical Takeaways from Figure 2:

1. **The Modulus Area Integrity Principle:** Area is a strictly non-negative physical quantity. When a curve dips below the x-axis, the negative definite integral must be negated to yield positive area.
2. **Orthogonal Parabola 16ab/3 Rule:** The region enclosed between two standard mutually orthogonal parabolas $y^2 = 4ax$ and $x^2 = 4by$ is always equal to $\frac{16ab}{3}$ sq units. For $y^2 = 4x$ and $x^2 = 4y$, $a = 1, b = 1 \implies \text{Area} = \frac{16}{3}$.
3. **Parabola & Line Shortcut:** The area enclosed by the parabola $y^2 = 4ax$ and the line $y = mx$ through the origin is given by $\frac{8a^2}{3m^3}$.

6. High-Speed CUET Shortcuts for Area Calculations

To finish the 50-question CUET Mathematics paper within 60 minutes, aspirants must deploy proven analytical shortcuts for standard conic sections and closed regions:

Geometric Region	Standard Equation(s)	Direct Area Formula (sq units)
Standard Circle	$x^2 + y^2 = r^2$	πr^2 (First quadrant $= \frac{1}{4}\pi r^2$)
Standard Ellipse	$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$	πab (First quadrant $= \frac{1}{4}\pi ab$)
Parabola & Latus Rectum	$y^2 = 4ax$ and $x = a$	$\frac{8}{3}a^2$
Two Orthogonal Parabolas	$y^2 = 4ax$ and $x^2 = 4by$	$\frac{16}{3}ab$
Parabola & Secant Line	$y^2 = 4ax$ and $y = mx$	$\frac{8a^2}{3m^3}$
Single Loop of Sine/Cosine	$y = \sin x, x \in [0, \pi]$ or $y = \cos x, x \in [-\frac{\pi}{2}, \frac{\pi}{2}]$	2 sq units
Between $y = \sin x$ and $y = \cos x$	$x \in [\frac{\pi}{4}, 5\frac{\pi}{4}]$	$2\sqrt{2}$ sq units
Parabola & Double Ordinate	$y^2 = 4ax$ from $x = 0$ to $x = x_0$	$\frac{4}{3}x_0y_0 = \frac{2}{3} \cdot$ (Bounding Rectangle)

ARCHIMEDES' QUADRATURE OF THE PARABOLA

Archimedes' classical theorem proves that the area bounded by any parabolic arc and a chord is exactly **two-thirds ($\frac{2}{3}$)** of the area of the circumscribed parallelogram/rectangle sharing the same base and height:

$$\text{Area} = \frac{2}{3} \cdot (\text{Base}) \cdot (\text{Height})$$

For $y^2 = 4ax$ bounded by $x = a$ (latus rectum):

- Base (latus rectum length) $= 4a$. Height $= a$.
- Area $= \frac{2}{3} \cdot (4a) \cdot (a) = \frac{8a^2}{3}$ sq units!

Chapter Synthesis & Cognitive Mindmap

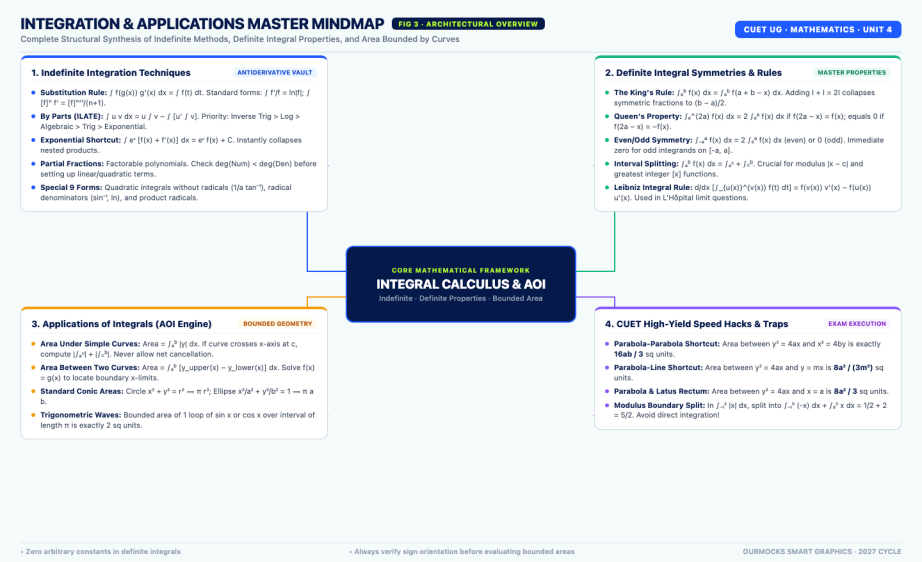


Figure 3: Figure 3: Integral Calculus Architectural Mindmap — Synthesis of Indefinite Methods, Definite Integral Properties, and Area Bounded by Curves.

HIGH-YIELD MEMORY HOOKS FOR CUET 2027

Keep these fundamental integration triggers primed:

- **Radical Inversion:** $\int \frac{dx}{\sqrt{a^2-x^2}} = \sin^{-1}\left(\frac{x}{a}\right) + C$. Remember: there is NO $\frac{1}{a}$ outside!
- **Quadratic Sum:** $\int \frac{dx}{x^2+a^2} = \frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right) + C$. Always includes $\frac{1}{a}$!
- **Exponential Collapse:** $\int e^x[f(x) + f'(x)]dx = e^x f(x) + C$.
- **King's Rule (P_4):** $\int_a^b f(x)dx = \int_a^b f(a+b-x)dx$. If integrand is $\frac{f(x)}{f(x)+f(a+b-x)}$, integral is simply $\frac{b-a}{2}$.
- **Symmetry Null:** $\int_{-a}^{a} f(x)dx = 0$ for any odd function ($f(-x) = -f(x)$).
- **Parabola Envelope:** Area bounded by $y^2 = 4ax$ and $x^2 = 4by$ is exactly $\frac{16}{3}ab$.

MASTER FORMULA & DECISION CHEATSHEET — INTEGRAL CALCULUS

1. Standard Radical & Quadratic Forms:

$$\int \frac{dx}{\sqrt{a^2-x^2}} = \sin^{-1}\left(\frac{x}{a}\right)+C, \qquad \int \frac{dx}{x^2+a^2} = \frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right)+C, \qquad \int \frac{dx}{\sqrt{x^2\pm a^2}} = \ln|x+\sqrt{x^2\pm a^2}|+C$$

2. Master Product Quadratic Forms:

$$\int \sqrt{a^2-x^2}dx = \frac{x}{2}\sqrt{a^2-x^2}+\frac{a^2}{2}\sin^{-1}\left(\frac{x}{a}\right)+C, \qquad \int \sqrt{x^2+a^2}dx = \frac{x}{2}\sqrt{x^2+a^2}+\frac{a^2}{2}\ln|x+\sqrt{x^2+a^2}|+C$$

3. By Parts & Exponential Shortcuts:

$$\int uv'dx = uv - \int u'vdx \quad \text{(ILATE priority)}, \qquad \int e^{kx}[kf(x)+f'(x)]dx = e^{kx}f(x)+C$$

4. Definite Integral Properties & Symmetries:

$$\int_a^b f(x)dx = \int_a^b f(a+b-x)dx \quad \text{(King's Rule)}, \qquad \int_0^{2a} f(x)dx = 2\int_0^a f(x)dx \quad (\text{if } f(2a-x)=f(x))$$

$$\int_{-a}^{a} f(x)dx = \begin{cases} 0 & \text{if } f(-x)=-f(x) \quad \text{(odd)} \\ 2\int_0^a f(x)dx & \text{if } f(-x)=f(x) \quad \text{(even)} \end{cases}, \qquad \frac{d}{dx}\left[\int_{u(x)}^{v(x)} f(t)dt\right] = f(v(x))v'(x)-f(u(x))u'(x)$$

5. Area Bounded by Curves Shortcuts:

Parabolas: $\text{Area}(y^2=4ax, x^2=4by) = \frac{16}{3}ab$, Line & Parabola: $\text{Area}(y^2=4ax, y=mx) = \frac{8a^2}{3m^3}$, Ellipse: πab

THE SEVEN FATAL CUET TRAPS IN INTEGRATION

1. **Inventing $\frac{1}{a}$ in Inverse Sine:** Writing $\frac{1}{a}\sin^{-1}\left(\frac{x}{a}\right)$ instead of $\sin^{-1}\left(\frac{x}{a}\right)$. Confusing $\sin^{-1}$ with $\frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right)$ is NTA's most frequent distractor.
2. **Modulus Integration Without Split:** Evaluating $\int_a^b |x-c|\,dx$ directly without partitioning at the root $x=c$. You must write $\int_a^c (c-x)dx + \int_c^b (x-c)dx$.
3. **Signed Definite Integral vs Geometric Area:** Allowing negative area below the x-axis to cancel positive area above. Total bounded area requires $|\int f(x)dx|$ across separate segments.
4. **Improper Rational Forms in Partial Fractions:** Failing to perform polynomial long division when $\deg(P) \geq \deg(Q)$ before setting up partial fraction fractions.
5. **Chain Rule Factor Error in Exponential Shortcut:** In $\int e^{kx}[kf(x)+f'(x)]dx = e^{kx}f(x)+C$, forgetting that k must multiply $f(x)$, not $f'(x)$.
6. **Omitting Limit Transformation in Substitutions:** Transforming $x \to t$ without updating the integration limits to $t_1 = g(a)$ and $t_2 = g(b)$.
7. **Intersection Strip Mismatch in AOI:** Integrating with respect to x when curves overlap vertically. Choosing horizontal strips (dy) avoids piecewise curve splitting.

Practice Questions (Exam Pattern)

Fifteen authentic CUET UG examination-standard multiple-choice questions testing conceptual, procedural, and computational mastery of integral calculus. Marking scheme: +5 for correct answer, −1 for incorrect attempt.

Part 1: Indefinite Integration & Algebraic/Trigonometric Methods

Q1. The indefinite integral $\int \frac{x^2}{1+x^6}dx$ evaluates to:

(A) $\frac{1}{2}\tan^{-1}(x^3)+C$ **(B)** $\frac{1}{3}\tan^{-1}(x^3)+C$
(C) $\frac{1}{3}\ln|1+x^6|+C$ **(D)** $\tan^{-1}(x^3)+C$

Sol. Q1. Let $t=x^3$, then $dt=3x^2dx$, giving $x^2dx=\frac{dt}{3}$. The integral becomes $\frac{1}{3}\int\frac{dt}{1+t^2}=\frac{1}{3}\tan^{-1}(t)+C=\frac{1}{3}\tan^{-1}(x^3)+C$. **Answer: (B)**

Q2. The value of the indefinite integral $\int\frac{e^{2x}-1}{e^{2x}+1}dx$ is equal to:

(A) $\frac{1}{2}\ln(e^{2x}+1)+C$ **(B)** $2\ln(e^x+e^{-x})+C$
(C) $\ln(e^x+e^{-x})+C$ **(D)** $\ln(e^{2x}+1)-x+C$

Sol. Q2. Divide numerator and denominator by e^x: $\int\frac{e^x-e^{-x}}{e^x+e^{-x}}dx$. Let $t=e^x+e^{-x}$, so $dt=(e^x-e^{-x})dx$. The integral becomes $\int\frac{dt}{t}=\ln|t|+C=\ln(e^x+e^{-x})+C$. Matches Option C. **Answer: (C)**

Q3. The evaluation of $\int\frac{dx}{\sqrt{9-25x^2}}$ yields:

(A) $\frac{1}{3}\sin^{-1}\left(\frac{5x}{3}\right)+C$ **(B)** $\frac{1}{5}\sin^{-1}\left(\frac{5x}{3}\right)+C$
(C) $\frac{1}{15}\sin^{-1}\left(\frac{5x}{3}\right)+C$ **(D)** $\sin^{-1}\left(\frac{5x}{3}\right)+C$

Sol. Q3. Factor out 25 from the radical: $\int\frac{dx}{5\sqrt{\left(\frac{3}{5}\right)^2-x^2}}=\frac{1}{5}\sin^{-1}\left(\frac{x}{\frac{3}{5}}\right)+C=\frac{1}{5}\sin^{-1}\left(\frac{5x}{3}\right)+C$. **Answer: (B)**

Q4. The antiderivative $\int e^x\left(\frac{1+\sin x}{1+\cos x}\right)dx$ simplifies to:

(A) $e^x\tan(x)+C$ **(B)** $e^x\sec^2\left(\frac{x}{2}\right)+C$
(C) $\frac{1}{2}e^x\tan\left(\frac{x}{2}\right)+C$ **(D)** $e^x\tan\left(\frac{x}{2}\right)+C$

Sol. Q4. Express using half-angle formulas: $\frac{1+2\sin\left(\frac{x}{2}\right)\cos\left(\frac{x}{2}\right)}{2\cos^2\left(\frac{x}{2}\right)}=\frac{1}{2}\sec^2\left(\frac{x}{2}\right)+\tan\left(\frac{x}{2}\right)$. Here $f(x)=\tan\left(\frac{x}{2}\right)$ and $f'(x)=\frac{1}{2}\sec^2\left(\frac{x}{2}\right)$. By $\int e^x[f(x)+f'(x)]dx=e^xf(x)+C$, we obtain $e^x\tan\left(\frac{x}{2}\right)+C$. Matches Option D. **Answer: (D)**

Q5. The value of the integral $\int x^2e^{3x}dx$ is given by:

(A) $\frac{e^{3x}}{27}(9x^2-6x+2)+C$ **(B)** $\frac{e^{3x}}{9}(3x^2-2x+2)+C$
(C) $\frac{e^{3x}}{27}(9x^2+6x+2)+C$ **(D)** $\frac{e^{3x}}{3}(x^2-2x)+C$

Sol. Q5. Using repeated integration by parts (tabular method): signs are $+-+-$, derivatives of x^2 are $x^2, 2x, 2, 0$, and integrals of e^{3x} are $\frac{e^{3x}}{3}, \frac{e^{3x}}{9}, \frac{e^{3x}}{27}$. Summing products: $\frac{x^2e^{3x}}{3}-\frac{2xe^{3x}}{9}+\frac{2e^{3x}}{27}+C=\frac{e^{3x}}{27}(9x^2-6x+2)+C$. **Answer: (A)**

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Part 2: Partial Fractions & Definite Integrals Properties

Q6. The decomposition of $\int \frac{x}{(x-1)(x-2)}dx$ results in:

(A) $\ln|\frac{x-1}{x-2}| + C$ **(B)** $\ln|(x-1)(x-2)| + C$
(C) $2\ln|x-2| - \ln|x-1| + C$ **(D)** $\ln|x-1| - 2\ln|x-2| + C$

Sol. Q6. By partial fractions: $\frac{x}{(x-1)(x-2)} = \frac{A}{x-1} + \frac{B}{x-2}$. Using cover-up rule: $A = \frac{1}{1-2} = -1$, and $B = \frac{2}{2-1} = 2$. Integrating gives $-\ln|x-1| + 2\ln|x-2| + C = 2\ln|x-2| - \ln|x-1| + C$. **Answer: (C)**

Q7. The value of the definite integral $\int_0^{\frac{\pi}{2}} \frac{\sqrt{\sin x}}{\sqrt{\sin x}+\sqrt{\cos x}}dx$ is:

(A) π **(B)** $\frac{\pi}{2}$
(C) 0 **(D)** $\frac{\pi}{4}$

Sol. Q7. Let $I = \int_0^{\frac{\pi}{2}} \frac{\sqrt{\sin x}}{\sqrt{\sin x}+\sqrt{\cos x}}dx$. Applying King's Property (P_4): $I = \int_0^{\frac{\pi}{2}} \frac{\sqrt{\cos x}}{\sqrt{\cos x}+\sqrt{\sin x}}dx$. Adding both equations: $2I = \int_0^{\frac{\pi}{2}} 1dx = \frac{\pi}{2} \to I = \frac{\pi}{4}$. Matches Option D. **Answer: (D)**

Q8. The value of $\int_{-1}^{1}(x^3 + x\cos x + \tan^5 x + 1)dx$ is:

(A) 0 **(B)** 2
(C) 1 **(D)** 4

Sol. Q8. The functions x^3, $x\cos x$, and $\tan^5 x$ are all odd functions ($f(-x) = -f(x)$). By P_7, their integrals over the symmetric interval $[-1, 1]$ evaluate to zero. The remaining integral is $\int_{-1}^{1} 1dx = [x]_{-1}^{1} = 1 - (-1) = 2$. **Answer: (B)**

Q9. The definite integral $\int_0^4 |x-2|\ dx$ evaluates to:

(A) 2 **(B)** 4
(C) 8 **(D)** 0

Sol. Q9. The integrand $|x-2|$ changes sign at $x = 2$. Partition the integral: $\int_0^2 (2-x)dx + \int_2^4 (x-2)dx$. First part: $\left[2x - \frac{x^2}{2}\right]_0^2 = 4 - 2 = 2$. Second part: $\left[\frac{x^2}{2} - 2x\right]_2^4 = (8-8) - (2-4) = 2$. Total area $= 2 + 2 = 4$. **Answer: (B)**

Q10. The value of $\int_0^{\pi} \frac{x\sin x}{1+\cos^2 x}dx$ is:

(A) $\frac{\pi^2}{4}$ **(B)** $\frac{\pi^2}{2}$
(C) $\frac{\pi}{4}$ **(D)** π^2

Sol. Q10. Let $I = \int_0^{\pi} \frac{x\sin x}{1+\cos^2 x}dx$. Using P_4: $I = \int_0^{\pi} \frac{(\pi-x)\sin x}{1+\cos^2 x}dx$. Adding: $2I = \pi\int_0^{\pi} \frac{\sin x}{1+\cos^2 x}dx$. Let $t = \cos x, dt = -\sin x dx$. When $x = 0 \to t = 1$, $x = \pi \to t = -1$. Then $2I = \pi\int_{-1}^{1} \frac{dt}{1+t^2} = \pi[\tan^{-1} t]_{-1}^{1} = \pi[\frac{\pi}{4} - (-\frac{\pi}{4})] = \frac{\pi^2}{2} \to I = \frac{\pi^2}{4}$. **Answer: (A)**

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Part 3: Advanced Definite Integrals & Area Bounded by Curves

Q11. If $f(x) = \int_0^x t\sin t\,dt$, then the derivative $f'(x)$ is:

(A) $\sin x + x\cos x$ **(B)** $x^2\sin x$
(C) $\cos x - x\sin x$ **(D)** $x\sin x$

Sol. Q11. By the Fundamental Theorem of Calculus (Leibniz Rule): $\frac{d}{dx}\left[\int_0^{v(x)} g(t)dt\right] = g(v(x))\cdot v'(x)$. Here $v(x) = x$, so $v'(x) = 1$, and $g(t) = t\sin t$. Therefore $f'(x) = x\sin x\cdot(1) = x\sin x$. Matches Option D. **Answer: (D)**

Q12. The area bounded by the curve $y = \cos x$ between $x = 0$ and $x = 2\pi$ and the x-axis is:

(A) 4 sq units **(B)** 0 sq units
(C) 2 sq units **(D)** 1 sq unit

Sol. Q12. Total bounded geometric area is strictly positive. The cosine curve has four symmetric quarter-period segments of area 1 sq unit each: from 0 to $\frac{\pi}{2}$ (above), $\frac{\pi}{2}$ to $3\frac{\pi}{2}$ (below, area 2), and $3\frac{\pi}{2}$ to 2π (above, area 1). Total area $= 1 + |-2| + 1 = 4$ sq units. (Note: the algebraic integral evaluates to 0, but geometric area is 4). **Answer: (A)**

Q13. The area enclosed between the two parabolas $y^2 = 4x$ and $x^2 = 4y$ is equal to:

(A) $\frac{8}{3}$ sq units **(B)** $\frac{32}{3}$ sq units
(C) $\frac{16}{3}$ sq units **(D)** 4 sq units

Sol. Q13. Here $4a = 4 \to a = 1$ and $4b = 4 \to b = 1$. Using the standard formula for orthogonal parabolas $y^2 = 4ax$ and $x^2 = 4by$: Area $= \frac{16}{3}ab = \frac{16}{3}\cdot(1)\cdot(1) = \frac{16}{3}$ sq units. **Answer: (C)**

Q14. The area of the region bounded by the parabola $y^2 = 8x$ and its latus rectum is:

(A) $\frac{16}{3}$ sq units **(B)** 8 sq units
(C) $\frac{32}{3}$ sq units **(D)** $\frac{64}{3}$ sq units

Sol. Q14. For $y^2 = 4ax$, $4a = 8 \to a = 2$. The latus rectum is the vertical line $x = a = 2$. The area bounded is $2\int_0^2 \sqrt{8x}dx = 2\cdot 2\sqrt{2}\left[\frac{x^{\frac{3}{2}}}{\frac{3}{2}}\right]_0^2 = 4\sqrt{2}\cdot\frac{2}{3}\cdot\left(2\sqrt{2}\right) = \frac{32}{3}$ sq units. Alternatively, using Archimedes formula: $\frac{2}{3}\cdot(\text{base})\cdot(\text{height}) = \frac{2}{3}\cdot(4a)\cdot(a) = \frac{8a^2}{3} = \frac{8\cdot 4}{3} = \frac{32}{3}$ sq units. **Answer: (C)**

Q15. The total area enclosed by the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$ is:

(A) 12π sq units **(B)** 25π sq units
(C) 7π sq units **(D)** 144π sq units

Sol. Q15. For the standard ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, semi-major axis $a = \sqrt{16} = 4$ and semi-minor axis $b = \sqrt{9} = 3$. The total enclosed area is $\pi ab = \pi\cdot 4\cdot 3 = 12\pi$ sq units. **Answer: (A)**

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